1. Basic Ping Scan

nmap -sn scanme.nmap.org

```
┌──(annie_20㉿kali)-[~]
└─$ nmap -sn scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:20 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.37s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds
```

Checks if host is alive (no port scanning)
Uses ICMP echo + ARP

2. TCP SYN Scan (Top 100 Ports)
nmap --top-ports 100 -sS scanme.nmap.org

```
┌──(annie_20㉿kali)-[~]
└─$ nmap --top-ports 100 -sS scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:20 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 94 closed tcp ports (reset)
PORT     STATE    SERVICE
22/tcp   open     ssh
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
445/tcp  filtered microsoft-ds
1026/tcp filtered LSA-or-nterm

Nmap done: 1 IP address (1 host up) scanned in 4.63 seconds
```

SYN scan sends SYN packets and analyzes responses without completing the TCP handshake, making it faster and stealthier.

3. Service Version Detection (Port 80)
nmap -p 80 -sV scanme.nmap.org

```
┌──(annie_20㉿kali)-[~]
└─$ nmap -p 80 -sV scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:20 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.36s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT   STATE SERVICE VERSION
80/tcp open  http    Apache httpd 2.4.7 ((Ubuntu))

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.46 seconds
```

Detects service running on port 80
Identifies version (e.g., Apache, nginx)
This helps identify software versions, useful for vulnerability detection.

4. OS Detection Scan
sudo nmap -O scanme.nmap.org

```
┌──(annie_20㉿kali)-[~]
└─$ sudo nmap -O scanme.nmap.org
[sudo] password for annie_20:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:20 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.24s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    open     http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
1022/tcp  filtered exp2
1023/tcp  filtered netvenuechat
1026/tcp  filtered LSA-or-nterm
9898/tcp  filtered monkeycom
9929/tcp  open     nping-echo
31337/tcp open     Elite
Aggressive OS guesses: Linux 4.15 (91%), Linux 4.19 - 5.15 (91%), Linux 5.4 (91%), IPFire 2.27 (Linux 5.15 - 6.1) (90%), Linux 2.6.32 (89%), Linux 4.15 - 5.19 (87%), Android T
V OS 11 (Linux 4.19) (87%), Linux 2.6.32 or 3.10 (87%), Linux 2.6.39 (87%), Linux 3.10 - 3.12 (87%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 14 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 40.56 seconds
```

Attempts to identify OS (Linux, Windows, etc.)
Uses TCP/IP fingerprinting to guess the target operating system.

5. Default NSE Scripts Scan
nmap -sC scanme.nmap.org

```
┌──(annie_20㉿kali)-[~]
└─$ nmap -sC scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 21:31 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 closed tcp ports (reset)
PORT      STATE    SERVICE
22/tcp    open     ssh
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)
80/tcp    open     http
|_http-title: Go ahead and ScanMe!
|_http-favicon: Nmap Project
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
1022/tcp  filtered exp2
1023/tcp  filtered netvenuechat
1026/tcp  filtered LSA-or-nterm
9898/tcp  filtered monkeycom
9929/tcp  open     nping-echo
31337/tcp open     Elite

Nmap done: 1 IP address (1 host up) scanned in 31.50 seconds
```

Runs default safe scripts
Checks for common vulnerabilities and info